

Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech
Code: 100306

Subject: Electrical Circuit Analysis

Time: 03 Hours
Full Marks: 70

Instructions: -

- (i) The marks are indicated in the right-hand margin.
- (ii) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer from the following (Any seven question only):

[2 x 7 = 14]

- (a) Thevenin's and Norton's equivalents are:
 - (i) Different representations with different values
 - (ii) Identical in all circuits
 - (iii) Equivalent with same behavior at load terminals
 - (iv) Applicable only for AC circuits
- (b) Mesh analysis is preferred when:
 - (i) The number of meshes is more than nodes
 - (ii) The number of meshes is less than nodes
 - (iii) Dependent sources are present
 - (iv) Circuits are complex
- (c) The steady-state response in a circuit is observed:
 - (i) During initial $t = 0$
 - (ii) When $t \rightarrow \infty$
 - (iii) When energy is maximum
 - (iv) Just after switching
- (d) In a critically damped RLC circuit, the roots of characteristic equation are:
 - (i) Complex conjugates
 - (ii) Real and distinct
 - (iii) Real and equal
 - (iv) Purely imaginary
- (e) The admittance of an inductor is:
 - (i) $j\omega L$
 - (ii) $1/(j\omega L)$
 - (iii) $-j\omega L$
 - (iv) $j\omega C$
- (f) The average power in a purely capacitive circuit is:
 - (i) Zero
 - (ii) Maximum
 - (iii) Depends on frequency
 - (iv) Infinite
- (g) Poles are values of 's' where:
 - (i) Input is zero
 - (ii) Output is zero
 - (iii) Denominator of transfer function is zero
 - (iv) Numerator is zero
- (h) Convolution in time domain becomes.....in Laplace domain:
 - (i) Division
 - (ii) Addition
 - (iii) Multiplication
 - (iv) Integration
- (i) A network is reciprocal if it contains only:
 - (i) Resistors
 - (ii) Resistors and capacitors
 - (iii) Passive linear elements
 - (iv) Active devices
- (j) The h-parameters are mainly used for:
 - (i) Filter design
 - (ii) Inductor modeling
 - (iii) Transistor modeling
 - (iv) Transformer design

- Q.2 (a) State Maximum Power Transfer Theorem. Prove that efficiency of the circuit under maximum power transfer condition is 50%. [7]
- (b) Find the voltage across 3Ω resistor using any theorem for the circuit shown in Figure-1 [7]

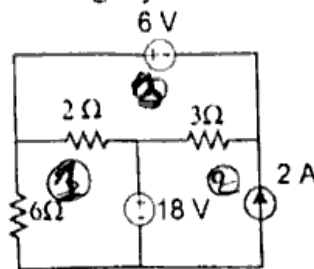


Figure 1

- Q.3 (a) Find V_A using the principle of superposition in network as shown in Figure 2. [7]

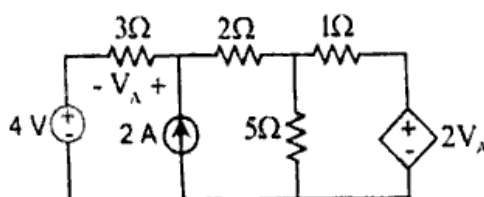


Figure 2

- (b) State Reciprocity Theorem. Verify the Reciprocity Theorem for the circuit shown in Figure 3. [7]

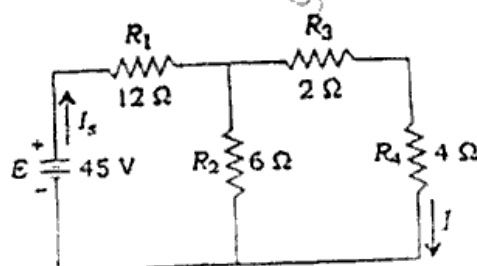


Figure 3

- Q.4 (a) Explain the behaviour of RL and RC elements for transient condition. Mention their representation at the time of switching. [7]
- (b) Draw the circuit of parallel RLC circuit and derive the formulae for resonant frequency, half-power frequencies, bandwidth and quality factor. [7]
- Q.5 (a) Determine Laplace transform of a following (i) $\sin 2t$ (ii) $\cos 2t$ [7]
- (b) Synthesis the waveform shown in Figure 4 below and Determine Laplace transform of periodic waveform. [7]

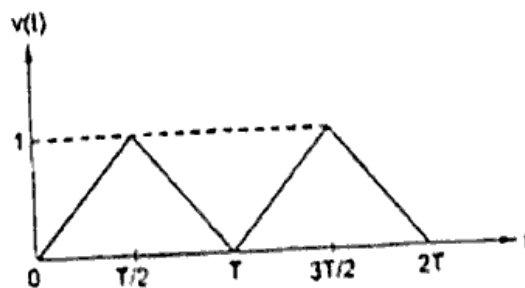


Figure 4

- Q.6 (a) To find Thevenin equivalent between terminals 'a' and 'b' as shown in Figure 5. [7]

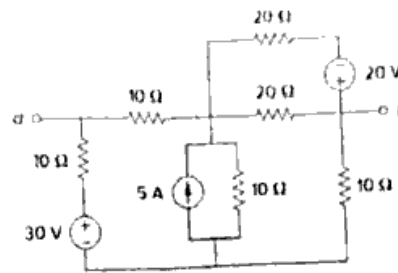


Figure 5

- (b) A switch is moved from position 1 to 2 at $t=0$ as shown in Figure 6. Find $V_R(t)$ and $V_C(t)$ for $t>0$. [7]

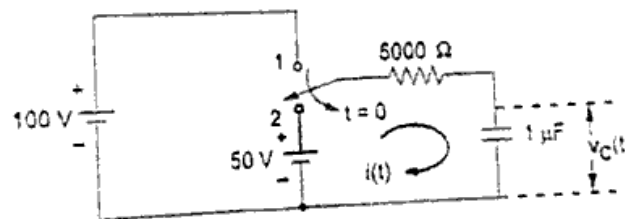


Figure 6

- Q.7 (a) Explain about the transmission parameters and derive the condition for symmetry and reciprocity. [7]
 (b) Determine z-parameter of network shown below in Figure 7 & give its equivalent circuit. [7]

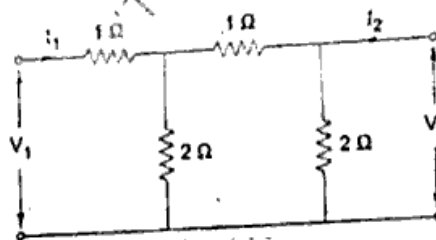


Figure 7

- Q.8 (a) Define two-port network and write assumptions to be made to find network and also obtain Y parameter. [7]
 (b) Two coils connected in series have an equivalent inductance of 10H. When the connections of one coil are reversed, the effective inductance is 6H. If the co-efficient of coupling is 0.6, calculate the self-inductance of each coil and mutual inductance. [7]

- Q.9 Write short notes on any two of the following: [7 x 2 = 14]
 (a) Poles and Zeros
 (b) Convolution integral
 (c) Mutual coupled circuits
 (d) Transmission and Hybrid parameters